

ratex-parser has unbounded parser recursion that leads to stack overflow (process abort)

Report Type: Weekly · Generated: July 08, 2026 01:52 UTC · Coverage: Jul 07 - Jul 08, 2026

1 Total Advisories	0 Critical	0 High	1 Medium
MEDIUM Severity	5.5 CVSS / Risk Score	TLP: CLEAR TLP Classification	PRO Customer Tier

Executive Summary

Summary RaTeX's recursive-descent parser recurses one (or more) native stack frame per nesting level at `{`, `\left`, `\sqrt{}`, `^`, etc, with **no** maximum depth limit. A short, ~10 KB input of nested groups overflows the 8 MB main-thread stack and aborts the process. With `panic = "abort"` (`Cargo.toml:48`), and because a Rust stack overflow is always a fatal `SIGABRT` regardless of panic strategy this is an unrecoverable, whole-process denial of service reachable from a single untrusted LaTeX string. Details The mutual recursion has no depth guard (`crates/ratex-parser/src/parser.rs`): `parse_expression (:113) -> parse_atom (:281/285) -> parse_group (:451) ^ | | on '{' (:459) recurse | +-----+` `\left` adds another recursive edge: `handle_left` `-> parse_expression` (`crates/ratex-parser/src/functions/left_rig`

CONFIDENTIALITY NOTICE: This report is classified PRO tier and intended exclusively for the named recipient. Redistribution, resale, or disclosure to unauthorized parties is strictly prohibited. © 2026 CYBERDUDEBIVASH SENTINEL APEX. All rights reserved.

Top Threat Advisories

High-priority advisories ranked by CVSS score, exploitability, and actor attribution.

Advisory Title	Severity	CVSS	EPSS	AI Summary
ratex-parser has unbounded parser recursion that leads to DoS attack over 15	MEDIUM	7.5	-	### Summary RaTeX's recursive-descent parser recurses one (or more) native s

MITRE ATT&CK® v15 Mapping

Technique density score: 0.0 · Techniques observed: 0 · Tactics covered: 0

MITRE ATT&CK techniques pending analyst enrichment. Deploy detection rules from Section 6 to identify related activity.

Threat Actor Intelligence

Attribution analysis across advisories in this report period.

Threat Actor	Count	Associated CVEs / Indicators
CDB-UNATTR-CVE	1	-

Indicators of Compromise (IOCs)

IOC extraction for this advisory is pending enrichment pipeline completion. PRO subscribers receive real-time IOC push to SIEM/SOAR via the SENTINEL APEX webhook at: **GET /api/v1/intell/iocs?advisory={id}**. Full IOC packages include IPv4, domain, SHA256, and URL indicators with confidence scores, first-seen timestamps, and STIX 2.1 formatted bundles.

CYBERDUDEBIVASH® SENTINEL APEX // CONFIDENTIAL - FOR AUTHORIZED USE ONLY

Detection Engineering

Deploy these production-ready detection rules into your SIEM platform to identify exploitation attempts in real time. Compatible with Microsoft Sentinel, Splunk ES, and any Sigma-compatible platform.

Sigma Rule - Webserver / Process Monitoring

```
title: SENTINEL APEX - Heap Overflow Exploitation Attempt
id: cdb-cdb-advisory-det
status: experimental
description: |
  Detects exploitation attempts targeting CDB-ADVISORY (heap-based overflow).
  Monitors for anomalous process crashes and memory allocation failures.
references:
  - https://intel.cyberdudebivash.com
author: CyberDudeBivash SENTINEL APEX
date: 2026/07/08
tags:
  - attack.initial_access
  - attack.t1190
  - attack.execution
  - attack.t1203
logsource:
category: process_creation
product: windows
detection:
  selection_crash:
    EventID: 1000
    ApplicationName|contains|any:
      - 'ratex-parser'
  selection_wer:
    EventID: 1001
    AppPath|contains|any:
      - 'ratex-parser'
  condition: selection_crash OR selection_wer
falsepositives:
  - Legitimate software bugs / crash reporting
level: medium
```

Microsoft Sentinel - KQL Query

```
// SENTINEL APEX - ratex-parser has unbounded parser recursion that leads to stack overflow (process abort)
// Advisory | Microsoft Sentinel KQL
let time_window = ago(24h);
SecurityAlert
| where TimeGenerated > time_window
| where Description has "ratex-parser has unbounded parser recurs"
or AlertName has "ratex-parser has unbounded parser recurs"
or ExtendedProperties has "ratex-parser has unbounded parser recurs"
| extend Rule = "APEX-ADVISORY", Severity = "MEDIUM"
| project TimeGenerated, AlertName, AlertSeverity, Entities, Description, Rule, Severity
| order by TimeGenerated desc
```

CYBERDUDEBIVASH® SENTINEL APEX // CONFIDENTIAL - FOR AUTHORIZED USE ONLY

Incident Response Playbook

Phase 1 - 0-24 Hours (Containment)

1. Activate IR team; assign lead analyst and executive sponsor.
2. Isolate or WAF-shield ratex instances exposed to the internet.
3. Enable verbose access logging on all ratex endpoints immediately.
4. Pull last 72h of web server and application logs for forensic baselining.
5. Check SENTINEL APEX IOC feeds and SIEM alerts for exploitation hits in your environment.
6. Notify relevant internal stakeholders and legal / compliance team.

Phase 2 - 24-72 Hours (Eradication)

1. Apply vendor patch for ratex or implement recommended mitigation.
2. Deploy Sigma and KQL detection rules from Section 6 across all SIEM platforms.
3. Rotate all API keys, service account credentials, and session tokens.
4. Conduct forensic timeline reconstruction for potential compromise window.
5. Validate patch deployment across 100% of affected instances via asset inventory.

Phase 3 - 7 Days (Recovery & Hardening)

1. Conduct post-incident review and update incident response runbooks.
2. Threat hunt for lateral movement or persistence artifacts using MITRE ATT&CK mapping.
3. Update asset inventory with patched version information and closure evidence.
4. Submit findings to internal risk register and CISO dashboard.
5. Schedule follow-up vulnerability scan to confirm remediation completeness.

Financial Impact Analysis (FAIR Model)

\$150K - \$900K Breach Cost Range (FAIR)	\$380K IBM Cost of Breach 2025 Median	\$120K/day Business Interruption Cost	\$85K Regulatory Fine Exposure
--	---	---	--

Medium severity events require internal investigation and targeted customer notification. Average cost includes 340 hours of engineering time.

Remediation Cost Estimate: \$55K · Includes engineering time, IR retainer activation, forensic investigation, and customer notification costs.

Regulatory & Compliance Implications

Organizations processing this advisory must evaluate obligations under applicable regulatory frameworks. The table below maps this threat to relevant compliance requirements.

Framework	Reference	Obligation
PCI-DSS v4.0	Req. 6.3.3	Non-critical patches applied within 3 months
NIST CSF 2.0	ID.RA / PR.PS	Vulnerability assessment and platform security policies
ISO 27001:2022	A.8.8	Vulnerability management programme with documented remediation timelines

Actionable Recommendations

Prioritized defensive actions based on this period's intelligence.

1. Monitor ratex-parser has unbounded parser recursion that leads to stack overflow (process abort) for escalation.
2. Apply patches in next maintenance window.
3. Apply vendor patch for ratex-parser has unbounded parser recursion that leads to stack overflow (process abort) immediately - check NVD for official fix guidance.
4. Enable enhanced logging and alerting on all systems running affected software versions.
5. Deploy the Sigma detection rule from this report into your SIEM platform (Sentinel / Splunk / Elastic).
6. Audit all internet-facing instances of affected products and confirm patch deployment.
7. Conduct a threat hunt using MITRE ATT&CK techniques mapped in this advisory.

Appendix - Report Metadata & Legal

Field	Value
Report ID	intel--9101cb2b33e7c01d09e1ea74
Report Type	Weekly
Platform	CYBERDUDEBIVASH SENTINEL APEX v166.0 GOD-MODE
Generated At	2026-07-08T01:52:07.53186
Customer Tier	PRO
Customer Email	-
TLP	TLP:CLEAR
STIX Version	2.1
ATT&CK Version	v15

Legal & Usage Notice

This threat intelligence report is produced by CYBERDUDEBIVASH SENTINEL APEX, operated by Bivash Nath (GSTIN: 21ARKPN8270G1ZP). All intelligence is aggregated from public vulnerability databases (NVD, CVSS), threat feeds, and proprietary AI enrichment pipelines. This report is provided AS-IS for informational and defensive security purposes only. CYBERDUDEBIVASH accepts no liability for actions taken based on this intelligence. Redistribution, resale, or disclosure to unauthorized parties is strictly prohibited without written consent. For licensing inquiries: enterprise@cyberdudebivash.com

CYBERDUDEBIVASH® SENTINEL APEX // CONFIDENTIAL - FOR AUTHORIZED USE ONLY